Laiden Ziegler

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EDUCATION

North Carolina State University — *B.S. Computer Science* *Raleigh, NC* · Anticipated May 2028

- Computer Science Courses (CSC 116, CSC 216, CSC 226) | GPA: 3.51/4.00

TECHNICAL PROJECTS

Lazee App — *Swift, SwiftUI, App Development*

- Designed, built, and published Lazee, an iOS strength training tracker, to the App Store as the sole developer
- Built the full app in Swift and SwiftUI, from the workout data model through the logging and progress-tracking UI

Studeo — *Electron, React, TypeScript, SQLite*

- Built and published a desktop study app for tracking courses, assignments, lectures, and study sessions, shipping macOS and Windows installers through a GitHub Actions CI pipeline
- Created an Electron main renderer split over typed IPC, backed by SQLite with versioned migrations, and covered the data and logic layers with roughly 99% unit and integration tests

PackSeats — *Python, Flask, Linux, systemd*

- Built and self-host a Python service that watches NC State course sections and pushes a phone alert the moment a full class opens up, running 24/7 on a cloud VM under systemd
- Architected four entry points (CLI, web planner, Telegram bot, and watcher daemon) around one shared fetch/parse core, integrated through a lock-guarded JSON store with no database
- Made concurrent writes safe with atomic temp-file renames behind a file lock, since the web UI and bot mutate the same watch list

FluidSim2D — *JavaScript, C++, Embedded Systems*

- Built a real time FLIP fluid simulation from scratch, implementing the full particle to grid solver with tilt driven gravity, advection, and surface tension so droplets pool and break apart naturally
- Ported the physics to C++ to run on an ESP32-C3 super mini driving a circular led matrix on a custom PCB, designing the solver to run within microcontroller memory and compute limits

Coursework Projects (CSC 216/217) — *Java*

- Built PackScheduler, a course registration system with FSM based input validation using the State pattern, custom linked list, and queue data structures, and role based login
- Applied object-oriented programming, JUnit unit and integration testing, and CI/CD with Jenkins

EXPERIENCE

Website/Ad Manager — *Alyce Leonard Permanent Makeup* *Lexington, NC* · Apr 2025 – Current

- Rebuilt the business's 13-page website from the ground up in HTML, CSS, and JavaScript, deployed on Cloudflare Pages
- Built a custom CMS on Cloudflare Pages Functions and D1 (SQLite) so a non-technical owner publishes posts and manages her gallery herself
- Secured the admin API by re-verifying Cloudflare Access JWTs inside the Worker, and wrote an allowlist HTML sanitizer to keep stored XSS out of the database

Sales Floor Associate / Cashier — *Gabe's* *Lexington, NC* · May 2023 – Apr 2025

- Strengthened problem-solving and multitasking abilities in a fast-paced retail environment
- Learned to work effectively as a team, having to coordinate with coworkers to maintain the store

TECHNICAL SKILLS

Languages: Java, C, Swift, Python

Tools/Software: Eclipse, VS Code, Vim, Xcode, Git, CAD, Microsoft Excel (Certified)

Concepts: Object-Oriented Programming, App Development, Unit Testing (JUnit), GUI Development, FSM's

HONORS

School Awards & Honors

- NC State Dean's List — May 2026
- National Rural and Small Town Recognition Program — Sept 2025